MILITARY SPECIFICATION

PUMP, CENTRIFUGAL, WATER, DIESEL-ENGINE-DRIVEN, 4-INCH, 600 GPM AT 50 FEET TOTAL HEAD

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers both skid-mounted and wheel-mounted, diesel-engine-driven, 4-inch, integral, self-priming, centrifugal pumps for water pumping service.

1.2 Classification. The pumps shall be of the following types, as specified (see 6.2):

Type I - Skid-mounted.
Type II - Wheel-mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-635 - Steel Plate, Carbon.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the se'f-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.

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MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13220E1090	- Pump, Centrifugal: Fresh Water, Diesel-Engine-Driven, Wheel-Mounted, Two Wheel Pneumatic Tires, 4-Inch Inlet, 4-Inch Outlet, Threaded, 600 GPM at 50 Ft. Head.
TA13220E1100	- Pump, Centrifugal: Fresh Water, Diesel-Engine-Driven, Skid-Mounted, 4-Inch Inlet, 4-Inch Outlet, Threaded, 600 GPM at 50 Ft. Head.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 48	- Gray Iron Castings.
A 126	- Gray Iron Castings for Valves, Flanges and Pipe Fittings.
A 276	- Stainless and Heat-Resisting Steel Bars and Shapes.
A 436	- Austenitic Gray Iron Castings.
B 26	- Aluminum-alloy Sand Castings.
B 148	- Aluminum-Bronze Sand Castings.
D 2000	- Elastomeric Materials for Automotive Applications.
D 3951	- Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

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(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017).

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 North West Seventh Steet, Miami, FL 33125.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

HYDRAULIC INSTITUTE (HI)

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump shall be in accordance with TA13220E1100 for type I and TA13220E1090 for type II, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit,

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assembly, and operation of the pump. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of pumps from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumps conforming to this specification.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and sub-components, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pump may be newly fabricated from recovered materials to the maximum extent practicable, provided the pump produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pump.

3.4 Screw-threads. All screw-threads shall conform to FED-STD-H28.

3.5 Environmental.

3.5.1 Operating temperature. The pump unit shall perform as specified herein in any ambient temperature from +32° F to +125° F.

3.5.2 Storage temperature. The pump unit shall not be damaged by storage in any ambient temperature from +160° F to -50° F.

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3.6 Safety. All shafts or rotating or moving parts of the pumps shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhaust or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges or corners are required or where they are not detrimental to safety.

3.6.1 Battery boxes and units. Batteries shall be securely held in place on carrier trays by means of locking devices and shall be protected against engine and exhaust heat. Battery compartments shall be ventilated and treated against acid fumes and spillage. The battery terminals shall be provided with guards to prevent accidental shorting. The battery circuit shall be provided with a reverse current device to protect the voltage regulator from damage when the batteries are connected in reverse polarity.

3.7 Human factors. The pumps shall conform to the human factors engineering criteria as specified in MIL-STD-1472.

3.7.1 Noise limits. The noise of the pumps shall not exceed the limits specified in MIL-STD-1474, category D for steady-state noise when tested in accordance with 4.5.2.6.1.1. The sound level shall comply with the exterior noise requirements of MIL-STD-1474 when tested in accordance with 4.5.2.6.1.2 (aural non-detectability and impulse noise do not apply). The precautions of MIL-STD-1474 (see 6.6) shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.5.2.6.1.3. Hearing protectors shall not be supplied with the pump unit. Hazard signs, when required, shall state: "CAUTION, HEARING PROTECTION REQUIRED WITHIN _____ FEET". Plates shall be in accordance with MIL-P-514, type III, composition C of type I, grade A, class 1 material.

3.8 Reliability. The specified mean-time-between-failure (MTBF) shall be 200 hours when the pump is operating at rated capacity and tested as specified in 4.5.2.7.

3.9 Maintainability.

3.9.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified herein (see 6.4 and 6.5), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.8.

3.9.2 Maintenance ratio. The pumps shall have a maintenance ratio of not more than 0.05. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating

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time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.10 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.11 Pump. The pump shall be of the centrifugal, single-stage, single-suction, top discharge, end-suction, integral self-priming type (after the pump case has been initially filled). The pump shall be designed so that the direction of discharge can be rotated 360 degrees, in mounting flange increments, on the horizontal plane. The pump shall be provided with a check valve, wear plate, and pump-case priming and drain ports, with threaded plugs. The wear plate shall be fabricated of steel conforming to QQ-S-635, SAE 1020. The suction and discharge ports of each pump unit shall be provided with flanges and threaded companion flanges. The companion flanges for both the suction and discharge ports shall be fabricated of gray iron castings conforming to ASTM A 48, class 30, or ASTM A 126, class B, and the flanges shall be internally threaded 4-inch NPT. The pump shall withstand an internal hydrostatic pressure equal to 1-1/2 times the shutoff head developed at maximum governed speed without leakage or permanent deformation when tested as specified in 4.5.2.11.

3.11.1 Pump case. The pump case shall be an aluminum casting conforming to ASTM B 26, Alloy 356.0-T6. The volute cut water shall be a gray iron alloy casting conforming to ASTM A 436, type 2, and shall be integrally cast into the pump case. The mounting dimensions of the case shall be such that it is compatible with or adaptable to the skid- and wheel-mounting frame hole spacing shown on the applicable drawings.

3.11.2 Pump shaft sleeve. The shaft sleeve shall be corrosion-resistant steel conforming to ASTM A 276, type 304, and shall be suitably sized to fit with both the impeller shaft and shaft seal. The pump shaft seal shall be mounted on the shaft sleeve.

3.11.3 Pump impeller. The pump impeller shall be fabricated of cast aluminum bronze conforming to ASTM B 148, Copper Alloy UNS No. C95500 and shall be of the single-suction, semi-open, non-clogging type. The impeller shall be designed to mount on the impeller shaft. The impeller shall be dynamically balanced to an out-of-balance condition of not more than 0.25 ounce-inch (oz-in) at 2600 revolutions per minute (rpm) maximum when tested as specified in 4.5.2.10.

3.11.4 Check valve. A check valve shall be provided on the suction inlet of each pump. The check valve shall be removable from the pump by removing the companion flange of the suction inlet. The valve shall prevent siphoning of water from the pump case. The check valve gasket and rubber components shall be made of rubber conforming to ASTM D 2000, type 2BG620E34 with nylon cloth insert.

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3.12 Diesel engine accessories. The diesel engine shall be furnished with but not limited to the following accessories:

a. Throttle control assembly to permit manually adjusting and locking in place to provide any engine speed up to the engine manufacturers maximum recommended speed.
b. Muffler(s) as required to reduce exhaust originated noise when tested as described in 4.5.2.6.1. If the noise level exceeds 85 dB(A), hazard signs shall be attached to the pump and shall state "Caution, Hearing Protection Required". Hearing protectors shall not be supplied with the pump. The signs shall conform to the requirements of MIL-STD-1472 for letter size and color for hazard signs.
c. Air cleaner. The air cleaner shall be dry type.
d. Air cleaner indicator. An air cleaner indicator shall be furnished to show when the air cleaner is restricted and needs cleaning.

3.12.1 Starting priming system. When specified (see 6.2), a cold starting aid conforming to MIL-E-52649 shall be furnished.

3.12.2 Fuel tank. A fuel tank shall be furnished. The capacity of the tank shall be sufficient for not less than 4 hours of continuous operation at the rated load. The tank shall be equipped with a cap and strainer. The tank shall be provided with a hand-operated fuel shutoff valve attached directly to the tank and a mechanically actuated fuel gage mounted on the tank. The fuel tank outlet shall be not less than 1/2 inch above the bottom of the tank. The tank outlet or the fuel line shall be provided with a 3-way selector valve and a 1/2-inch x 20 SAE flared fitting for connection with an auxiliary fuel supply. A corrosion-resisting circular metal face plate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK", and "AUXILIARY".

3.13 Performance characteristics.

3.13.1 Priming. Prior to operation, the pump case shall be filled with water to eliminate air from the pump chamber and to aid in initial priming of the pump.

3.13.2 Performance. When operated at conditions equivalent to standard sea level atmospheric conditions of 29.92 inches of mercury and water with a specific weight of 62.3 pounds per cubic foot at a temperature of 68° F, using 15 feet or 30 feet of vertical suction line, as applicable, and one elbow, the pump shall perform as follows:

a. Prime and deliver its rated capacity of 600 gallons per minute (gpm) at 50 feet total head in not more than 3 minutes when operated under a 10-foot static suction lift.
b. Prime and deliver not less than 250 gpm in not more than 4 minutes when operated under a 25-foot static suction lift.

The suction and discharge pipes or hoses and fittings shall be of the same nominal size as the suction and discharge ports of the pump.

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10 minutes of the initial attempt and shall demonstrate the ... then be started within

3.13.3 Head capacity. The pump shall deliver not less than the capacities stated below and as shown in figure 1:

a. 300 gpm at a 110-foot head.
b. 450 gpm at a 100-foot head.
c. 600 gpm at a 50-foot head.

When operating the pump unit at rated speed and at 600 gpm at a 50-foot head, the net positive suction head (NPSH) shall not exceed 12.5 feet. At 24 rpm, the pump unit shall equal or exceed this performance. In addition, the pump shall require not more than the continuous duty engine brake horsepower (bhp) at any operating condition shown in figure 1.

3.13.4 Pump efficiency. The pump shall develop an efficiency of not less than 65 percent while delivering 450 gpm at rated speed.

3.14 Electromagnetic interference. The electromagnetic interference emission characteristics of the pumps shall conform to MIL-STD-461, class IIIC.

3.15 Lubrication.

3.15.1 Lubricants. The pumps shall operate as specified herein when serviced with the following military lubricants, grades as applicable to the operating conditions:

Engine:	MIL-L-2104 or MIL-L-46167.
Grease lubrication:	MIL-G-10924 or MIL-G-23827.

3.15.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3451, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extenison. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.15.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.16.1), with 1 percent copper 8-quinolinolate or 6 percent salicylanilide (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

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b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with South Coast Air Quality Management District, Rule 442.

3.17 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the pumps to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209.

3.18 Identification marking. The pumps shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pumps on plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.18.1 Identification plates. Each pump shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the pump. Instruction plates shall conform to MIL-STD-514, type III, composition C of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.19 Treatment and painting. The portions of the pumps and components or parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.20 Workmanship.

3.20.1 Castings and forgings. All parts, components, and assemblies of the pumps which include castings and forgings shall be clean of harmful extraneous materials such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.20.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means

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to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.20.3 Welders and welding.

3.20.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall make available to the contracting officer certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractor's who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.20.3.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the pumps to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pumps shall be at the option of the contractor.

3.20.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

3.20.5 Connections, hoses, piping, and cables. All connections, hoses, piping, and cables shall be supported to prevent damage caused by vibration, chafing, or sharp bends.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract,

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the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7 and 6.8).

4.1.2 Disassembly inspection. Failure of any first article examination or test shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article (see 4.3).
b. Quality conformance inspection (see 4.4 and 4.7.1).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article pumps shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first article pumps shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level III.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. AQL shall be 2.5 percent defective.

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4.4.3 Tests.

4.4.3.1 Individual. Each pump shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pumps shall be examined as specified herein for the following major defects:

101. Any part (or component) not in accordance with the QAP requirements as shown on the drawings.
102. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified.
103. Fabrication not in accordance with referenced drawings.
104. Material not as specified.
105. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
107. Contractor does not have documentation available for identification of material, material finishes or treatments.
108. Components or parts missing or not as specified.
109. Electromagnetic interference suppression components and network not identical to the approved model.
110. Assembly of components incorrect or bolted connection not securely tightened.
111. Screw-threads not conforming to FED-STD-H28.
112. Battery boxes and units not as specified.
113. Identification marking missing or not legible.
114. Identification plates not as specified.
115. Treatment and painting not as specified.
116. Hazard sign missing or not as specified.
117. Pump case not as specified.
118. Pump shaft sleeve not as specified.
119. Pump impeller not as specified.
120. Check valve not as specified.
121. Engine accessories not as specified.
122. Starting priming system not as specified.
123. Fuel tank not as specified.
124. Lubricants not as specified.
125. Lubrication fittings not as specified.
126. Pressure-release device not as specified.
127. Electrical circuitry not protected from the effects of moisture and fungus growth.

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128. Workmanship not as specified.
129. Castings and forgings not as specified.
130. Metal fabrication not as specified.
131. Welding not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to testing, the pump shall be lubricated with grease and oils as specified in 3.15.1. Tests shall be performed in accordance with the applicable provisions of Test Code for Centrifugal Pumps of the Hydraulic Institute. Unless otherwise specified herein, the tests shall be conducted using a 10-foot static suction lift. Water shall be used as the test liquid, and the static lift shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury. The data shall be corrected to the specific weight of water (62.3 pounds per cubic foot) at a temperature of 68° F. A data sheet indicating pump efficiency, head capacity, and brake horsepower required by the pump shall be supplied with the first article pumps.

4.5.2.2 Test schedule.

TABLE I. Test schedule.

First article	Quality Conformance		Tests	Test paragraph	Requirement paragraph
	Individual	Sample			
1	2	3	4	5	6
X			High temperature	4.5.2.3	3.5
X			Low temperature	4.5.2.4	3.5
X			Safety	4.5.2.5	3.6
X			Human factors	4.5.2.6	3.7
X			Reliability	4.5.2.7	3.8
X			Maintainability	4.5.2.8	3.9
X			Pump brake hp	4.5.2.9	3.13.3
X	X		Impeller balance	4.5.2.10	3.11.3
X	X		Hydrostatic pressure	4.5.2.11	3.11
X	X		10 ft. prime performance	4.5.2.12.1a	3.13.2
X		X	25 ft. prime performance	4.5.2.12.1b	3.13.2
X		X	Head capacity	4.5.2.12.2	3.13.3
X	X		Fuel tank	4.5.2.13	3.12.2
X			Electromagnetic interference	4.5.2.14	3.14
X			Tiedown provisions	4.5.2.15	3.17

4.5.2.3 High temperature. The pump shall be tested as specified in MIL-STD-810, procedure I, method 501. The maximum ambient storage temperature

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shall be 160° F, and the maximum ambient operating temperature shall be 125° F. The operating period of procedure I shall be 2 hours. The pump shall be operated at rated capacity. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.4 Low temperature. The pump unit shall be tested as specified in MIL-STD-810, procedure I, method 502. The minimum ambient storage temperature shall be minus 50° F. The minimum ambient operating temperature shall be plus 32° F. The contractor shall certify the suitable operation down to an ambient temperature of plus 32° F. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.5 Safety. The pumps shall be evaluated throughout testing as specified in 4.5.2. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.6 Human factors. The human factors shall be evaluated during testing. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.6.1 Noise level measurement.

4.5.2.6.1.1 Representative duty position. Measure the noise level at every 45 degrees, plus or minus 5 degrees, of azimuth, 65 inches, plus or minus 1 inch, above ground level, at a 25-foot radius from the geometric center of the equipment in accordance with MIL-STD-1474 for steady-state noise. The pumps shall be operated at rated load. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.6.1.2 Exterior. Measure the noise level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.7 shall constitute failure of this test.

4.5.2.6.1.3 Hazard precautions. Measure the noise level at all control and maintenance positions while operating at the maximum noise condition for those locations. Should the noise level exceed 85 dB(A) at any position, failure to provide the precautions specified in 3.7 shall constitute failure of this test.

4.5.2.7 Reliability. Using the MTBF specified in 3.8, the pumps shall be tested as specified in 4.5.2.7.1 and 4.5.2.12.2 with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan IV. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pump by continued operation.
c. Safety hazards to personnel.

Nonconformance to 3.8 shall constitute failure of this test.

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4.5.2.7.1 Reliability test profile. The following test profile shall be repeated until an "accept" or "reject" decison is achieved in accordance with the test procedures specified in 4.5.2.7. Prime and start the pump (see 3.13.1), and operate the pump in the following sequence:

a. Operate 3 hours at 300 gpm at a 110-foot total head.
b. Operate 3 hours at 450 gpm at a 100-foot total head.
c. Operate 4 hours at 600 gpm at a 50-foot total head.

4.5.2.8 Maintainability.

4.5.2.8.1 Maintenance support evaluation. Maintenance operations required during testing shall be accomplished to determine conformance to 3.9.1. Failure of the support elements to maintain the pump unit as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.8.2 Maintenance ratio. The maintenance ratio shall be computed during the reliability test specified in 4.5.2.7. Nonconformance to 3.9.2 shall constitute failure of this test.

4.5.2.9 Pump brake horsepower determination. Prior to assembly of the first article pump to the engine, operate the pump as specified in 3.13.3 utilizing a dynamometer or calibrated electric motor to determine the brake horsepower required by the pump. A required brake horsepower in excess of the continuous horsepower rating of the engine (see 3.13.3) or a peak pump efficiency of less than 65 percent shall constitue failure of this test.

4.5.2.10 Balancing of impeller and shaft. Prior to assembly, the pump impeller and shaft shall be tested for out-of-balance condition by a dynamometer at the operating speed of the pump. An out-of-balance condition of more than 0.10 oz-in shall constitute failure of this test.

4.5.2.11 Pump hydrostatic pressure. With check valve installed, subject the pump to an internal hydrostatic pressure equal to 1-1/2 times the shutoff head obtained at maximum operating speed. Leakage or permanent deformation of the pump shall constitute failure of this test.

4.5.2.12 Pump performance.

4.5.2.12.1 Priming. Prime the pump as specified in 3.13.1. After priming, operate the pump as follows:

a. At rated capacity (600 gpm at 50 feet total head) for a period of not less than 10 minutes using a 10-foot static suction lift.
b. At 250 gpm for a period of not less than 10 minutes using a 25-foot static suction lift.

Nonconformance to 3.13.2 or permanent deformantion of any component shall constitute failure of this test.

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4.5.2.12.2 Head capacity. Operate the pump to determine conformance to the head capacitites shown in figure 1. The operating time shall be that time requred to obtain the performance characteristics of the pump. Pump delivering less than capacities and heads shown in figure 1 shall constitute failure of this test.

4.5.2.13 Fuel tank. Apply an internal air pressure of 5 psig to the fuel tank with all openings plugged or capped and immerse in water so that the tank top is not less than 12 inches under the water level. Any evidence of leakage from the tank shall constitute failure of this test.

4.5.2.14 Electromagnetic interference. The first article shall be tested in accordance with MIL-STD-462 to determine conformance with 3.14. The testing agency shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the test report shall constitute failure of this test.

4.5.2.15 Tiedown provisions.

4.5.2.15.1 Test procedure. With the pump anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns.

4.5.2.15.2 Failure criteria. Weld failure, permanent deformation of the pump or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6 Inspection comparision. The Government may select pump units at any time during the contract production period and subject these pump units to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pump units selected at random from those which have been accepted by the Government and will not include the previously inspected first article pump units. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparision, and failure of such additional tests shall have the same effect as failure of those test specified as inspection comparison.

4.6.1 Inspection failure. Failure of any inspection comparison pump unit to meet any requirement specified herein during and as a result of the examination and test specified in 4.6 shall be cause for rejection of the inspection comparison pump unit and shall be cause for refusal by the Government to continue acceptance of production pump units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiences. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pump units previously accepted and produced under the contract. Any deficiencies found as a result of the inpsection comparison will

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be considered prima facie evidence that all pump units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of packaging.

4.7.1.1 Unit of product. For the purpose of inspection, a completely packed pump prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the applicable defects listed in MIL-P-10603. Additionally, each sample shall be examined for the following defects. The AQL shall be 2.5 percent defective.

No.	Defect	A	B	Comm
132.	Street-ell not removed from flange as specified.	5.1.1.1	5.1.2	
133.	Pump not presrved as specified.	5.1.1.2	5.1.2	5.1.3
134.	Wheel-mounted pumps not packed as specified.	5.2.1.1	5.2.2.1	5.2.3
135.	Skid-mounted pumps not packed as specified.	5.2.1.2	5.2.2.2	5.2.3
136.	Marking not as specified.	5.3.1	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. The degree of preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The street-ell supplied for the pump outlet shall be removed from the flange and the lifting bail complete with washers and nuts shall be removed. For skid-mounted pumps, only the street-ell shall be removed. The street-ell may be secured to the intermediate bearing housing with a tie of soft annealed wire.

5.1.1.2 Pump. The pump and all associated components shall be preserved in accordance with the level A requirements of MIL-P-10603.

5.1.2 Level B. Disassembly shall be as specified in 5.1.1.1 for level A. The pump and all associated components shall be preserved in accordance with the level B requirements of MIL-P-10603.

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5.1.3 Commercial. Each complete pump and associated components shall be preserved in accordance with the commercial requirements of MIL-P-10603 except reference therein to FED-STD-356 shall be corrected to reference ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Wheel-mounted pumps. The plywood cap for the wheel-mounted pumps shall be fabricated in accordance with PPP-B-601, overseas type, style A, grade B box, except that 1/4-inch plywood and 2-3/4-inch-width cleats shall be used (see figure 4). The cap shall fit against the side channels, rear reflectors, and pump inlet, and rest on the trailer tongue. Spacers of nominal 2- by 2-inch lumber cut 25 inches long shall be secured to the four inside vertical corners of the cap to properly position the cap on all sides. The cap shall be secured to the trailer by strapping over the box and under the trailer frame and sealing on top of the cap as shown in figure 4. The strapping shall be stapled to the cap as shown.

5.2.1.2 Skid-mounted pumps. The plywood cap for the skid-mounted pumps shall be fabricated as specified in 5.2.1.1, except that the filler cleats on the cap top shall be of nominal 4-inch material in lieu of the 2-3/4-inch-width filler cleates for the top of the wheel-mounted pumps. The cap shall fit over the pumps and be supported on the 2- by 2-inch spacer blocks. The pumps shall be strapped as shown in figure 2. Wood skids attached to the metal skid base shall consist of nominal 2- by 4-inch skids with nominal 1- by 4-inch rubbing strips and may be of one-piece construction with minimum thickness of 2-3/8 inches. Skids shall extend out from the metal frame 1 inch and shall be fastened to the frame of the unit with 3/8-inch carriage bolts of appropriate length fastened with a nut and lockwasher.

5.2.2 Level B.

5.2.2.1 Wheel mounted. The wheel-mounted pumps shall be covered with a cap fabricated of triple-wall corrugated fiberboard conforming to PPP-B-640, class 2, grade A, style RSC (without bottom) box, in accordance with figure 5.

5.2.2.2 Skid mounted. The skid-mounted pumps shall be covered with a cap as specified in 5.2.2.1, except the requirements of figure 3 shall apply.

5.2.3 Commercial. Each complete pump shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, shipping weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The pumps are intended for use in general-purpose, water-pumping operations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of pumps required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Time frame required for submission of first article, and number of pumps required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. Engine cold starting aid required (see 3.12.1)
g. Color required (see 3.19).
h. When the welding process will not be at the option of the contractor (see 3.20.3.2).
i. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item should be a preproduction pump. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pumps.

6.6 Noise limits. When the noise level of the pumps exceeds 85 dB(A) and technical infeasibility is established via MIL-STD-1474 (paragraph for maximum nosie limits), or the means of reduction is too costly, the written permission of the procuring activity shall be requested to exceed the 85 dB(A) limit. If

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the procedures of MIL-STD-1474 have been pursued and documented to the satisfaction of the procuring activity, then the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (manuals paragraph) shall be provided. The discussion will include the requirement for hearing protection, the noise level of the pumps, the distance from the pumps at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

6.7 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.8 Definition.

6.8.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

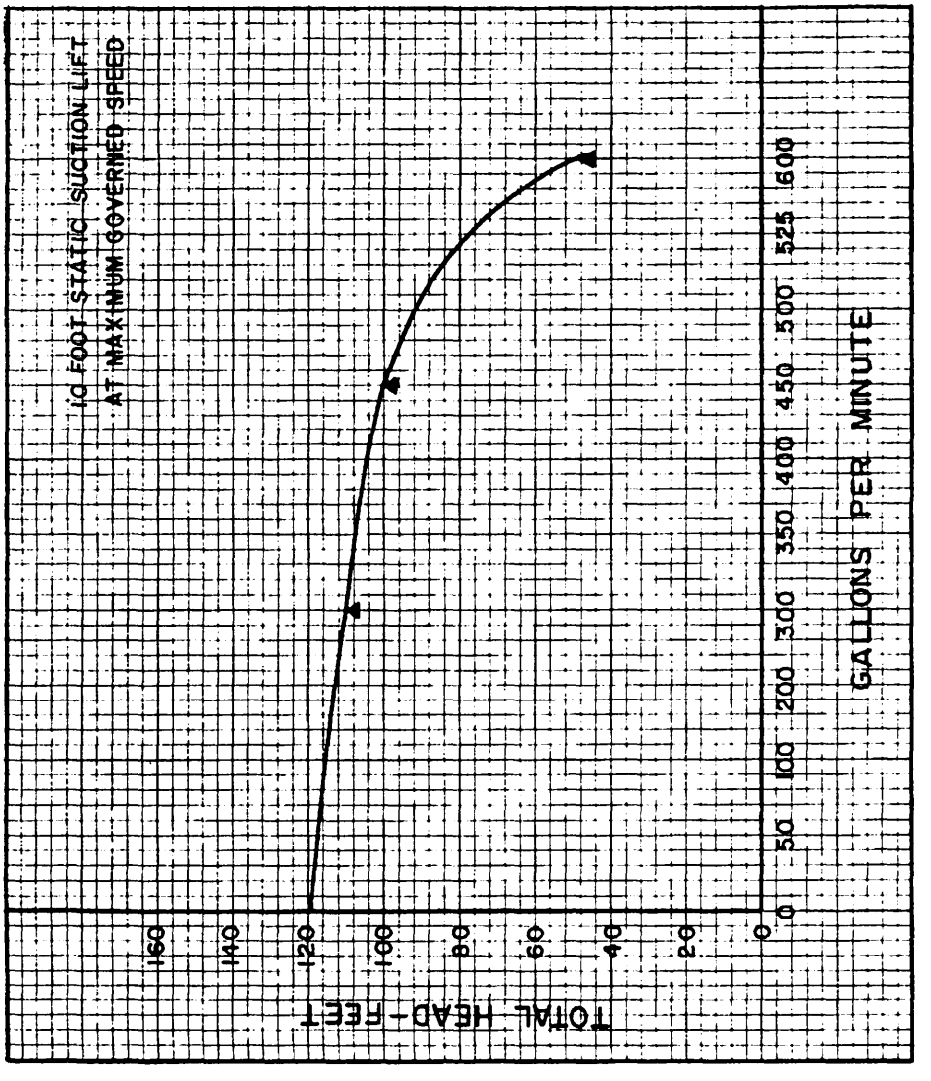
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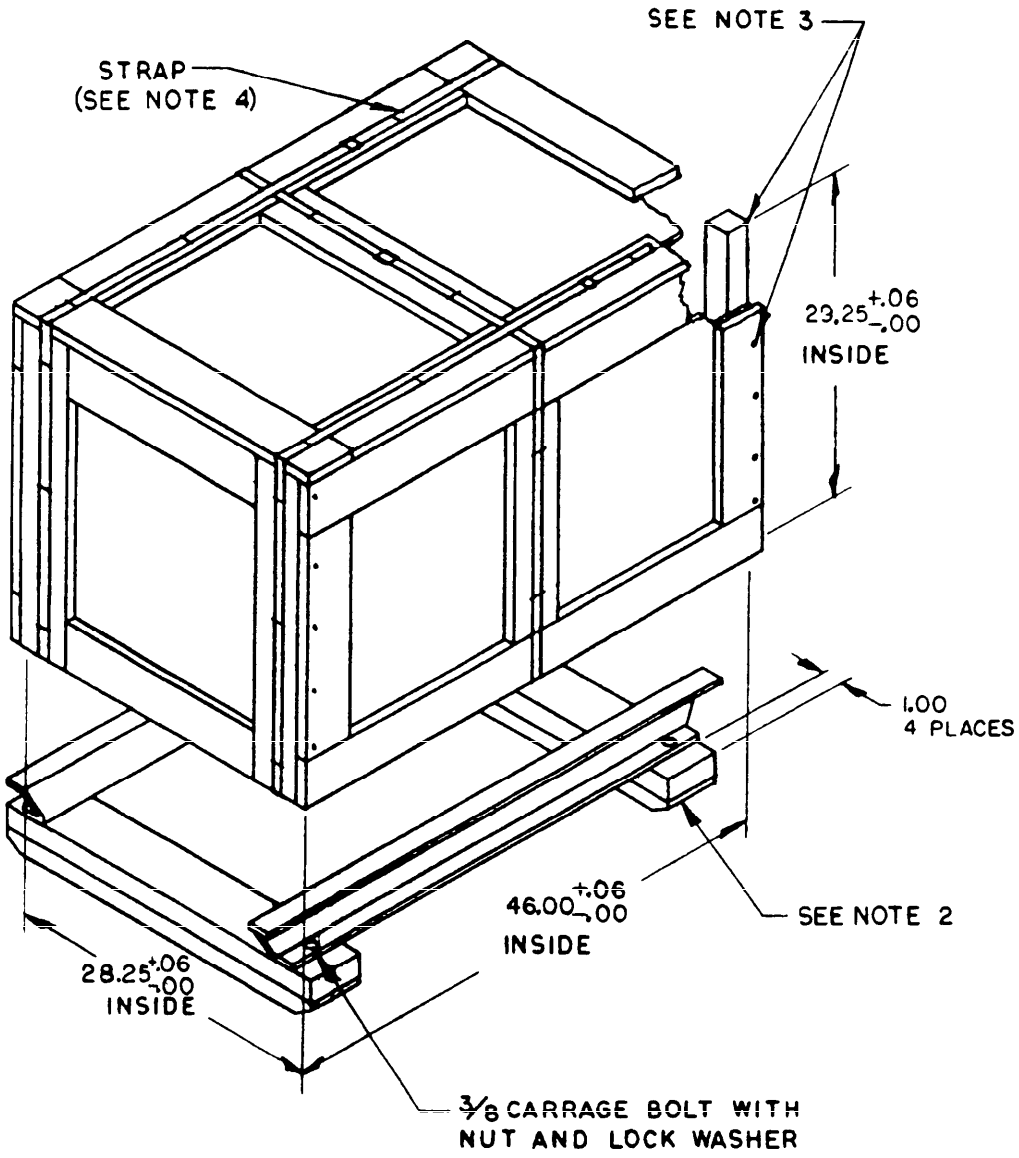
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X-351A

FIGURE 1. Head capacity.

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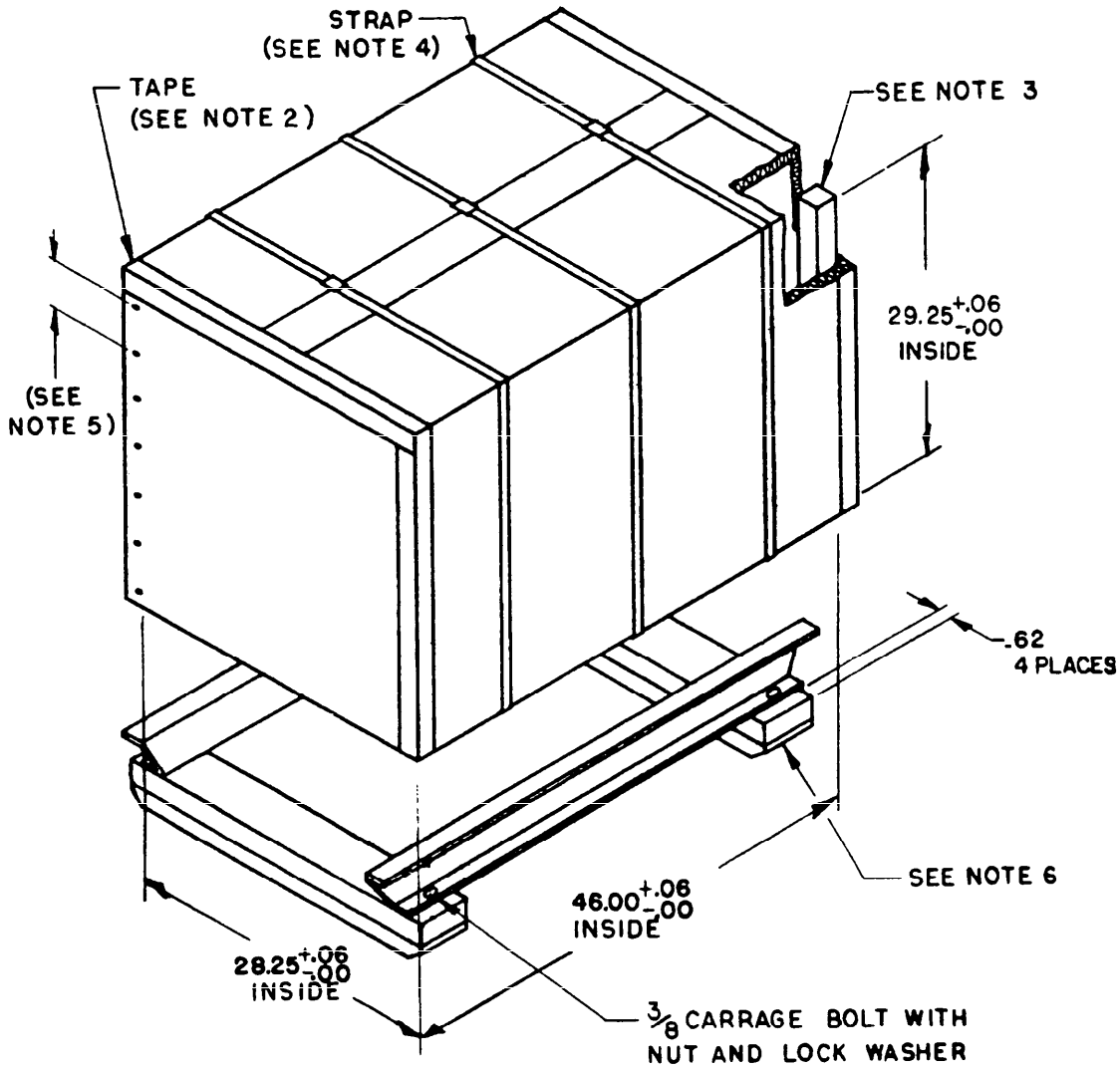
NOTES

1. CAP IN ACCORDANCE WITH PPP-B-601, WITH EXCEPTIONS, SEE 5.2.1.2.
2. SKID 2x4 NOM- RUBBING STRIP 1x4 NOM (OPTIONAL 2 3/8 x 4 MIN THK 1 PIECE CONSTRUCTION)
3. SPACERS, PER MIL-STD-731, CLASS 2, 2 x 2 NOM SIZE, CUT TO 25.00 LONG, 4 CORNERS, SECURE WITH 7d CEMENT COATED BOX NAILS
4. STRAP, PER QQ-S-781, TYPE I OR IV, FINISH A, CLASS 1, 3/4 X .020 THK, AND SEALS.
5. ALL DIMENSIONS ARE IN INCHES.

FIGURE 2. Packing level a-skid mounted unit.

X-248A

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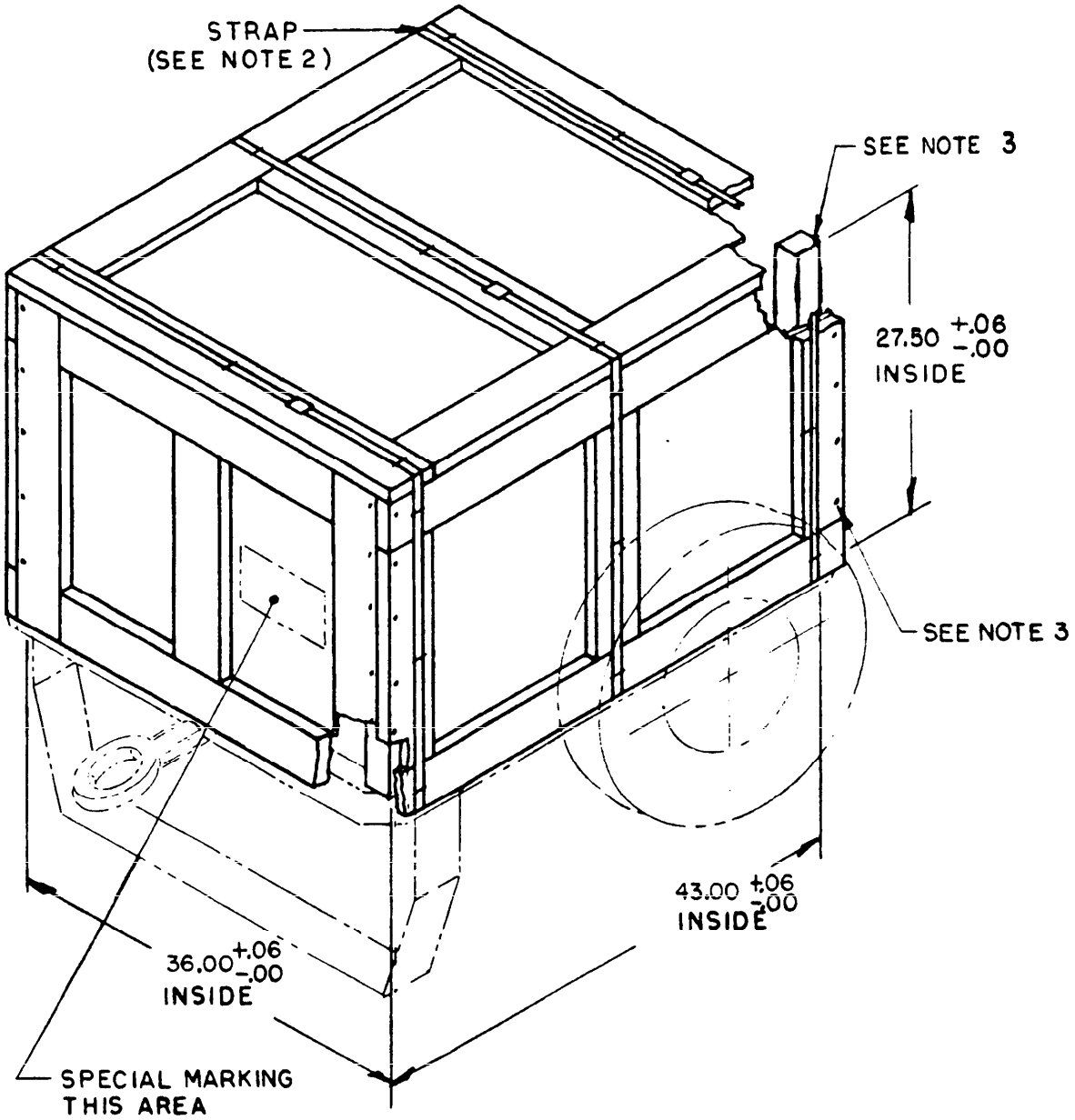
NOTES

1. CAP IN ACCORDANCE WITH PPP-B-640, WITH EXCEPTIONS, SEE 5.2.2.2.
2. TAPE, PER PPP-T-60, CLASS 1, 3 WIDE, COVER ALL JOINTS.
3. SPACERS, PER MIL-STD-731, CLASS 2, 2x2 NOM SIZE, CUT TO 25.00 LONG, 4 CORNERS (SECURE WITH NAILS—NOTE 5)
4. STRAP, PER QQ-S-781, TYPE I OR IV, FINISH A, CLASS 1, 3/4 X.020 THK, AND SEALS.
5. GALVANIZED ROOFING NAILS, 1/2 DIA HEAD x 1 1/4 LONG ON 4.00 CENTERS.
6. SKID 2x4 NOM-RUBBING STRIP 1x4 NOM (OPTIONAL: 2 3/8 MIN x 4 1 PIECE CONSTRUCTION)
7. ALL DIMENSIONS ARE IN INCHES

FIGURE 3. Packing level b-skid mounted unit.

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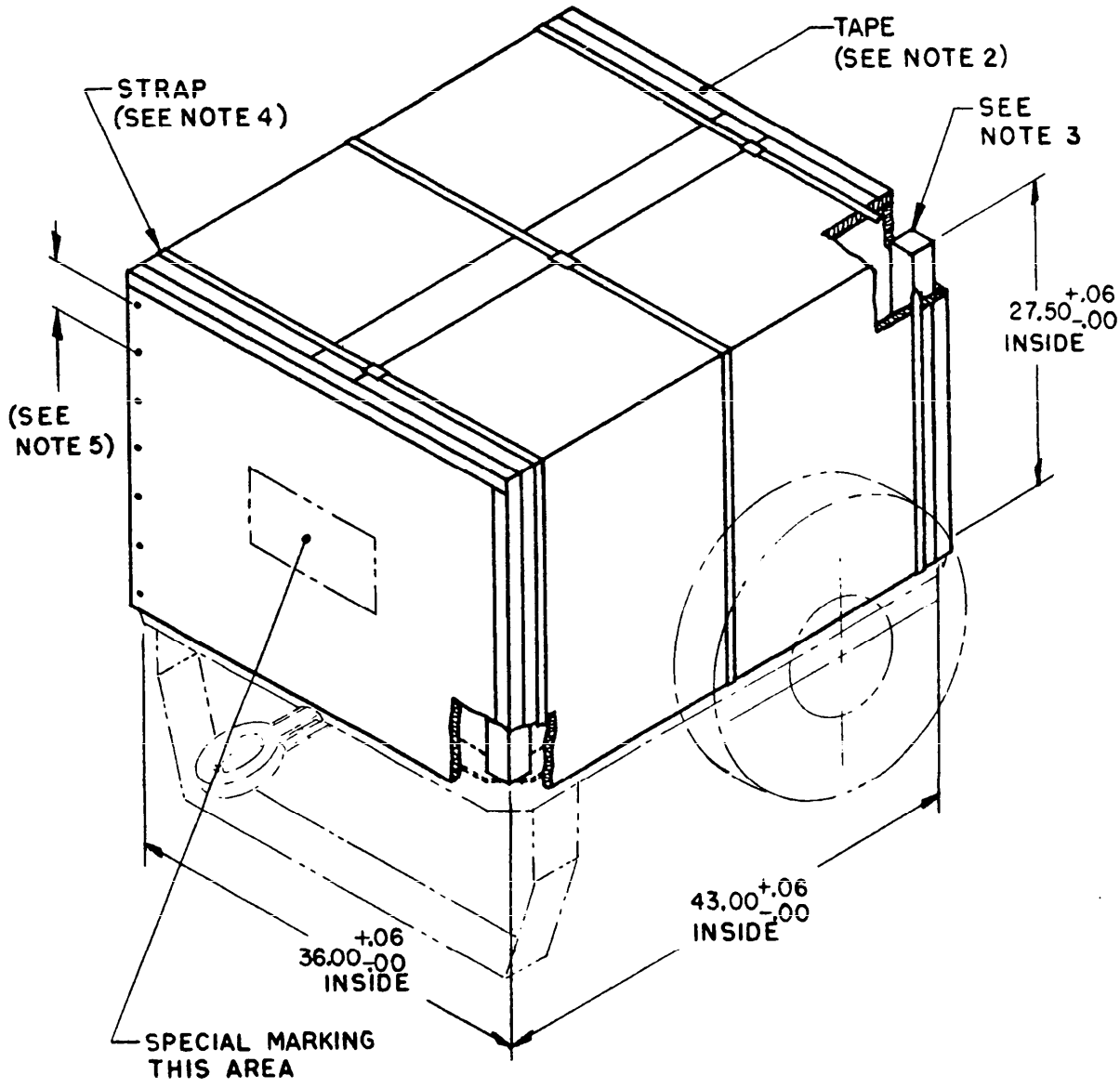
NOTES

1. CAP IN ACCORDANCE WITH PPP-B-601, WITH EXCEPTIONS. SEE 5.2.1.1.
2. STRAP, PER QQ-S-781, TYPE I OR IV, FINISH A, CLASS 1, 3/4 X.020 THK, AND SEALS.
3. SPACERS, PER MIL-STD-731, CLASS 2, 2 x 2 NOM SIZE, CUT TO 25.00 LONG, 4 CORNERS. SECURE WITH 7d CEMENT COATED BOX NAILS.
4. ALL DIMENSIONS ARE IN INCHES.

FIGURE 4. Packing level a-wheel mounted unit.

X-246A

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NOTES

1. CAP IN ACCORDANCE WITH PPP-B-640,WITH EXCEPTIONS, SEE 5.2.2.1
2. TAPE, PER PPP-T-60, TYPE III,CLASS 1, COVER ALL JOINTS. 3WIDE.
3. SPACERS, PER MIL-STD-731, CLASS 2, 2x2 NOM SIZE, CUT TO 25.00 LONG, 4 CORNERS. SECURE WITH NAILS-NOTE 5.
4. STRAP, PER QQ-S-781, TYPEI OR IV, FINISH A, CLASS I, 3/4 X.020 THK, AND SEALS.
5. GALVANIZED ROOFING NAILS, 1/2 DIA HEAD x 1 1/4 LONG ON 4.00 CENTERS.
6. ALL DIMENSIONS ARE IN INCHES

FIGURE 5. Packing level b - wheel mounted unit.

X-247A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-52474B(ME)	Pump, Centrifugal, Water, DED, 4-Inch, 600 GPM at 50 Feet Total Head

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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